

EE 102: Signal Processing and Linear Systems

Problem Set #6: Fourier Series**Name:** _____**Submission Date:** _____

Problem 1 Consider the RC circuit from Pre-requisite #3, with input voltage $x(t)$ and output voltage $y(t)$ measured across the capacitor. Use $R = 330\ \Omega$ and $C = 1000\ \mu\text{F}$, so $RC = 0.33\ \text{s}$. Its impulse response is

$$h(t) = \frac{1}{RC} e^{-t/(RC)} u(t).$$

For each of the input signals below, find the Fourier series representation of the **periodic steady-state output** $y(t)$.

(a) [20 points] A train of impulses:

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - n)$$

Hint: Start by finding the Fourier series representation of the input signal, $x(t)$. So, write

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t},$$

where $\omega_0 = 2\pi/T$ is the fundamental frequency and T is the period of the input signal. Find a_k using the Fourier series analysis equation. Then, use the property of LTI systems to find the output coefficients b_k (that is, the Fourier Series coefficients of the output signal) in terms of the input coefficients a_k .

$$b_k = a_k H(jk\omega_0),$$

where $H(j\omega)$ is the frequency response of the system. Refer to the lecture notes to find out the frequency response formula!

(b) [30 points] A square wave of amplitude 1 sketched below in Figure 1.

Hint: Over one period, $-\frac{1}{2} \leq t < \frac{1}{2}$, the signal is

$$x(t) = \begin{cases} 1, & -\frac{1}{4} \leq t \leq \frac{1}{4}, \\ 0, & \text{otherwise within this period.} \end{cases}$$

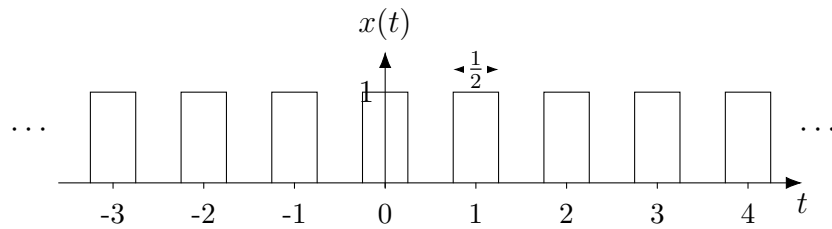


Figure 1: A square wave of amplitude 1, period $T = 1$, and duty cycle 50%.

(c) [20 points] Write Python/MATLAB code to implement the input signals above. Compute the outputs and their Fourier series representations, and plot the magnitude of the outputs, $|y(t)|$, on a graph.

Note that you will need to choose a finite number of harmonics to include in the Fourier series representation (computers cannot handle infinite sums!). Additionally, you will need to choose a sampling interval for your simulation.

You may use AI tools to get help with the syntax and to get a set up going – you must write your own Fourier series computation code though.

Problem 2 Compute the complex Fourier series representation of each signal below. For each signal $f(t)$, let c_k be its Fourier series coefficients and $\omega_0 = 2\pi/T_0$. Define the truncated representation by

$$f_N(t) = \sum_{k=-N}^N c_k e^{jk\omega_0 t}.$$

Thus, retain the DC term and the first N positive and negative harmonics. For $N = 1, 2, 3, 4$, compute the error energy **over one period**:

$$E_N = \int_0^{T_0} |f(t) - f_N(t)|^2 dt.$$

You may use computer programs for computations. Show how you obtain the coefficients and the error energy.

(a) [15 points] A sawtooth signal with fundamental period $T_0 = 1$. Figure 2(a) shows one period:

$$x(t) = t, \quad 0 \leq t < 1, \quad x(t+1) = x(t).$$

Extend this definition periodically to all positive and negative time.

Hint: Evaluate the coefficient integral over $0 \leq t < 1$. Integration by parts can be useful for the nonzero harmonics.

(b) [15 points] A rectangular signal with fundamental period $T_0 = 2$, shown in Figure 2(b). Over one period,

$$y(t) = \begin{cases} 1, & 0 \leq t < 1, \\ 0, & 1 \leq t < 2, \end{cases} \quad y(t+2) = y(t).$$

The signal repeats periodically for all positive and negative time.

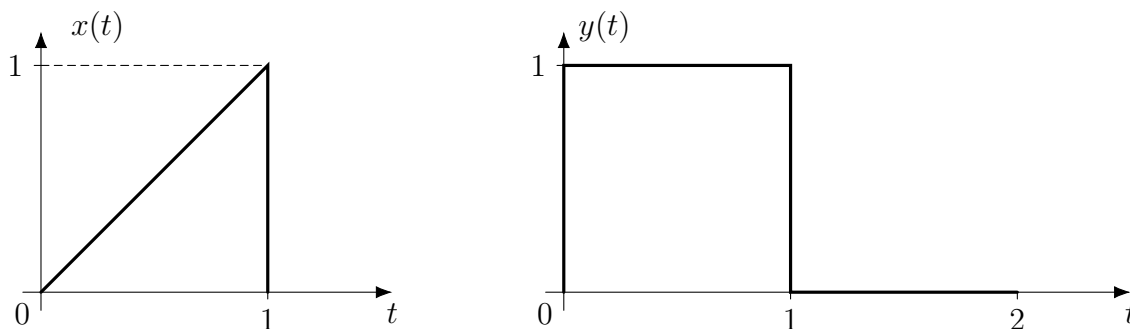


Figure 2: One period of (a) the sawtooth signal $x(t)$ and (b) the rectangular signal $y(t)$.